List of Publications

Major Contributions

Jan 2024 | *The infalling elliptical galaxy M89: the chemical composition of the AGN disturbed hot atmosphere*
Kara S., **Plšek T.**, Protušová K., Breuer J.-P., Werner N., Mernier F., Ercan E. N.: MNRAS, 528, 1500-1516

Nov 2023 | *CADET: pipeline for detection of X-ray cavities in hot galactic and cluster atmospheres*
Plšek T., Werner N., Topinka, M., Simionescu A.: MNRAS, 527, 3315-3346

Sep 2022 | *The relation between accretion rate and jet power in early-type galaxies with thermally unstable hot atmospheres*
Plšek T., Werner N., Grossová R., Topinka M., Simionescu A., Allen S. W.: MNRAS, 517, 3682-3710

Contributions

Jul 2026 | *IceCube AGN neutrino source candidate PKS 2233−148: peculiar observed properties & possible gravitational lensing signatures*
Britzen S., Kovačević A. B., Popović L. Č., Kun E., Zajaček M., Jaron F., **Plšek T.**, Pashchenko I. N., Pánis R., Stuchlík Z., Tursunov A.: MNRAS, 552, stag1355

Mar 2026 | *X-ray line diagnostics of the multi-phase gas in the Centaurus cluster core with XRISM/Resolve*
Kondo M., Fukushima K., Suda K., Majumder A., Sato K., Matsushita K., Mernier F., Nakazawa K., Simionescu A., Breuer J.-P., Fukazawa Y., Fujimoto R., Hatsukade I., Hosogi K., Loewenstein M., **Plšek T.**, Sun M., Urata M., Werner N., Yamasaki N. Y., Fujita Y.: PASJ, 78, 852-866

Feb 2026 | *Chemical composition and enrichment of the Centaurus cluster core seen by XRISM/Resolve*
Mernier F., Fukushima K., Simionescu A., Kondo M., Majumder A., **Plšek T.**, Werner N., Fujita Y., Sato K., Matsushita K., Loewenstein M., Mushotzky R., Breuer J.-P., Fujimoto R., Fukazawa Y., Hatsukade I., Nakazawa K., Urata M., Yamasaki N.: A&A, 706, A86

Nov 2025 | *The thermodynamic structure and large-scale structure filament in MACS J0717.5+3745*
Breuer J.-P., Werner N., **Plšek T.**, Mernier F., Umetsu K., Simionescu A., Devlin M., Di Mascolo L., Dibblee-Barkman T., Dicker S., Mason B. S., Mroczkowski T., Romero C., Sarazin C. L., Sievers J.: A&A, 703, A231

Oct 2025 | *Compact stellar systems hosting an intermediate-mass black hole: Magnetohydrodynamic study of inflow-outflow dynamics*
Labaj M., Ressler S. M., Zajaček M., **Plšek T.**, Ripperda B., Peißker F.: A&A, 702, A233

Jul 2025 | *X-ray investigation of the remarkable galaxy group Nest200047*
Majumder, A., Simionescu, A., **Plšek, T.**, Brienza, M., Churazov, E., Khabibullin, I., Gastaldello, F., Botteon, A., Röttgering, H., Brüggen, M., Lyskova, N., Rajpurohit, K., Sunyaev, R. A., Wise, M. W.: A&A, 699, A375

Feb 2025 | *MeerKAT discovery of gigahertz radio emission extending from Abell 3017 towards Abell 3016*
Hu D., Werner N., Xu H., Zheng Q., Breuer J.-P., Wu L., Duchesne S. W., van Weeren R. J., Sun M., Zhang C., Johnston-Hollitt M., Shan H., Guo Q., Zhu Z., Wang J., Gu J., Zhao Y., Siew H., Mao J., Zhang Z., **Plšek T.**: A&A, 694, A320

Feb 2025 | *The bulk motion of gas in the core of the Centaurus galaxy cluster*
XRISM Collaboration [including **Plšek T.**]: *Nature*, 638, 365-369

Nov 2024 | *ICECUBE AGN neutrino candidate PKS 1717+177: dark deflector bends nuclear jet*
Britzen S., Kovačević A. B., Zajaček M., Popović L. Č., Pashchenko I. N., Kun E., Pánis R., Jaron F., **Plšek T.**, Tursunov A., Stuchlík Z.: MNRAS, 535, 2742-2762

Oct 2023 | *Ram-pressure stripped radio tail and two ULXs in the spiral galaxy HCG 97b*
Hu D., Zajaček M., Werner N., Grossová R., Jáchym P., Roberts I. D., Ignesti A., Kenney J. D. P., **Plšek T.**, Breuer J.-P., Shimwell T., Tasse C., Zhu Z., Wu L.: MNRAS, 527, 1062-1080

Feb 2022 | *Very Large Array Radio Study of a Sample of Nearby X-Ray and Optically Bright Early-type Galaxies*
Grossová R., Werner N., Massaro F., Lakhchaura K., **Plšek T.**, Gabányi K., Rajpurohit K., Canning R. E. A., Nulsen P., O'Sullivan E., Allen S. W., Fabian A.: ApJS, 258, 30

Submitted Publications

2026 | *Missing metals in the core of the Centaurus cluster*
Plšek T., Majumder A., Werner N., Mernier F., Simionescu A., Štofanová L.: submitted to MNRAS

Nov 2025 | *IceCube candidate neutrino emitter PKS 1413+135 is gravitationally lensed on pc-scales*
Britzen S., Zajaček M., Fendt C., Best H., Jaron F., Pashchenko I.N., **Plšek T.**, and Seidel G.: submitted to MNRAS